

SO project - Stochastic Machine Replacement Problem

Yiping Fang, LGI, CentraleSupélec

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Problem statement

Basic problem

At the end of each production cycle (e.g., seasonal), a cheese production line must decide whether to continue using a machine or replace it with a new one. A machine in the production line at cycle t has a corresponding efficiency state $s_t \in S = \{1, 2, \dots, 10\}$ depending on the degradation level of the machine. A higher value of s_t represents a more degraded state and thus low productivity of the production line.

We know the machine's state at the first cycle $s_1 = 1$, and it has probability $p = 0.9$ to go to efficiency state $s_{t+1} = \min\{s_t + 1, 10\}$ and probability $1 - p$ to go to efficiency state $s_{t+1} = \min\{s_t + 2, 10\}$ if not replaced by a new one.

At each efficiency state $s \in S$, it produces $y(s) = 5 + s - 0.15s^2$ tons of cheese over the corresponding production cycle (e.g., at state $s = 10$, the productivity is $y(10) = 0$). The profit contribution of cheese production is $m = 150$ k€ per ton. We assume the machine can be replaced at any production cycle, and the net cost of replacing a machine is $c = 500$ k€ and we cannot produce any cheese during this production cycle due to this replacement activity.

This is an infinite-horizon stochastic model with time t measured in production cycles. The state is.

s_t = production efficiency state of a machine at production cycle t
 $s_t \in \{1, 2, \dots, 10\}$

The action is

$$a = \begin{cases} 0 & \text{do nothing} \\ 1 & \text{replace the machine} \end{cases}$$

The state transition function is

$$T(s_t, a, s_{t+1}) = \begin{cases} p & a = 0, s_{t+1} = s_t + 1, s_t \leq 8 \\ 1 - p & a = 0, s_{t+1} = s_t + 2, s_t \leq 8 \\ 1 & a = 0, s_{t+1} = 10, s_t = 9 \text{ or } 10 \\ 1 & a = 1, s_{t+1} = 1, \forall s_t \end{cases}$$

The reward function is

$$r(s, a) = \begin{cases} m \cdot y(s) & \text{if } a = 0 \\ -c & \text{if } a = 1 \end{cases}$$

where

$$y(s) = 5 + s - 0.15s^2$$

Questions:

- Write down the Bellman optimality equation of the value function.
- What replacement policy maximizes the expected long-term cumulative profit with discount factor γ ? (Solve the problem using value iteration)
- Test the sensitivity of optimal policies to the replacement cost c , the discount factor γ , and the state transition probabilities p .

A more different setting: multi-machine serial line and multi-line coordination (Optional)

In a real cheese factory (like a Gruyère or Comté facility), we often have multiple parallel production lines. Each line is a serial system consisting of multiple machines, and each line is only as fast as its slowest machine. Let's assume we have **two** production lines, and each of them is a sequence of **two** specialized machines:

- Machine A (pasteurizer/vat): handles the initial heating and curdling.
- Machine B (draining/molding): separates whey and shapes the cheese.

Now, in the multi-machine serial line problem, the factory state $\mathbf{s}$ is a vector of 4 integers: $\mathbf{s} = (s_{1,A}, s_{1,B}, s_{2,A}, s_{2,B})$. If any machine in line i degrades, the entire line slows down. The production output of line i at time t is determined by the higher state among its machine A and B, and the total production of the factory for a given state $\mathbf{s}_t$ is

$$y(\mathbf{s}) = y(\max\{s_{1,A}, s_{1,B}\}) + y(\max\{s_{2,A}, s_{2,B}\})$$

If one machine in a line hits state $s = 10$ (0 output), the entire line stops, regardless of how new the other machine is.

For the replacement decisions of the 4 machines, let $a_{i,j} \in \{0, 1\}$ be the decision to replace machine $j \in \{A, B\}$ in line $i \in \{1, 2\}$. However, there is **only one maintenance team** to cover the whole factory. We can only replace one machine in the entire factory per cycle, even if two are broken. This creates a hard constraint on the action space:

$$\sum_{i \in \{1, 2\}} \sum_{j \in \{A, B\}} a_{i,j} \leq 1 \Leftrightarrow \|\mathbf{a}\|_1 \leq 1$$

The joint reward function $\mathcal{R}(\mathbf{s}, \mathbf{a})$ accounts for the fact that maintenance on one line does not halt production on the other. If machine j in line i is replaced ($a_{i,j} = 1$), line i produces no cheese, but the other line continues to operate based on its current state. The cost c is incurred for any replacement activity:

$$\mathcal{R}(\mathbf{s}, \mathbf{a}) = \sum_{i \in \{1, 2\}} \left[m \cdot y(\max\{s_{i,A}, s_{i,B}\}) \cdot \left(1 - \sum_{j \in \{A, B\}} a_{i,j}\right) \right] - c \cdot \|\mathbf{a}\|_1$$

Furthermore, to ensure the factory meets its obligations to cheese distributors, management imposes a **global reliability constraint**: the factory must maintain an average expected production level of at least $y_{\min}$ (e.g., 70% of theoretical maximum capacity) over the infinite horizon.

The transition probability $P(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{a})$ is the product of individual transitions, assuming degradation is independent across machines:

$$P(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{a}) = \prod_{i \in \{1, 2\}} \prod_{j \in \{A, B\}} T(s_{i,j,t}, a_{i,j}, s_{i,j,t+1})$$

Questions

- Formulate the problem as a dual LP to handle the reliability constraint naturally. Hint: define the variables $x(\mathbf{s}, \mathbf{a})$ as the occupancy measures (the discounted probability of being in state $\mathbf{s}$ and taking action $\mathbf{a}$), and incorporate the reliability constraint based on the occupancy measures.
- Implement and solve the LP and identify if the optimal policy is randomized in certain states (e.g., “Replace with 40% probability”).

- Analyze the cost of reliability: How much does the total profit decrease for every 1% increase in the $y_{\min}$ requirement?

Submissions

To submit your work, please first access the link:

- <https://nextcloud.centralesupelec.fr/s/RLXSN9kgjfNDEWY>

In the folder there is already a *name_list.csv* file listing your names appeared in the Edunao system. After you finishing work in the provided `submission2.ipynb`, rename your notebook by

- `S0-P2-Firstname-Lastname.ipynb`

For example, for the first student it should be `S0-P2-Youssef-Abouelaz.ipynb`. Upload your file to this folder in the previous link. You can delete your previous submission and to upload a new one if modification is needed. Note that all activities are logged by the cloud server. So please try not to accidentally delete or download files of others.